

P&ID Review
Action Items List

Item	Requested By	Description	Associated P&ID	Action By	Requested Date	Required Date	Completion Date	Comments and Updates
001	P&ID Review	SSP to create a P&ID Review / Cover Sheet for the next issuance of P&IDs.	General comment	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
002	P&ID Review	Motiva to consider alternative location for sanitary sewer tie-in from new control room.	YC-169496	Motiva	12/17/2025			
003	P&ID Review	Motiva to consider "right sizing" of PC stormwater tank to accommodate additional first flush stormwater from boiler complex.	YC-199790	Motiva	12/17/2025			
004	P&ID Review	Motiva to consider increasing line size of potable water header to account for two safety showers operating simultaneously.	YC-199791	Motiva	12/17/2025			
005	P&ID Review	Motiva to consider increasing line size of plant and instrument air headers to allow for additional requirements of the boiler complex.	YC-199791	Motiva	12/17/2025			
006	P&ID Review	SSP to add backflow prevention for liquids from the fuel gas coalescer. Backflow prevention will be dual dissimilar check valves.	YC-199791	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
007	P&ID Review	Motiva to confirm philosophy for sweeping and purging of flare and impacts on nitrogen and natural gas usage.	YC-216703 YC-216710	Motiva	12/17/2025			
008	P&ID Review	Motiva to consider impact of fuel gas requirements for Boiler Complex combined with Aromatics plant and whether other users would be starved of fuel gas.	YC-216712	Motiva	12/17/2025			
009	P&ID Review	Motiva to consider direct discharge of first flush and non-oily wastewater (boiler blowdown, demin regenerant and cooling tower blowdown).	YC-216717	Motiva	12/17/2025			This would require an environmental permit modification.
010	P&ID Review	Motiva to consider the need for neutralization of demin regenerant (part of non-oily wastewater currently).	YC-216717	Motiva	12/17/2025			Motiva currently neutralizes demin regenerant. SSP experience is that demin regenerant should not require neutralization.
011	P&ID Review	SSP to move the clarified water line to a separate battery limit drawing to indicate that it is located at Res7.	YC-216983	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
012	P&ID Review	Motiva to consider adjustments to clarified water system design to resolve issues in existing system.	YC-216983	Motiva	12/17/2025			
013	P&ID Review	SSP to add a blind on the bleed side of each block valve for battery limit isolation.	Global	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
014	P&ID Review	SSP to add full battery limit instrumentation on slop oil line.	0000-PID-0010	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
015	P&ID Review	Motiva to consider separation of clarified water and redundancy of single point of supply.	Global	Motiva	12/17/2025			
016	P&ID Review	Motiva to consider additional sand filter to increase redundancy. Concerns were expressed about reliability and technology selection of sand filters.	1000-PID-0005	Motiva	12/17/2025			
017	P&ID Review	Motiva to consider customization of the sand filter equipment package.	1000-PID-0005	Motiva	12/17/2025			Current path forward is to defer until bid package is developed in the next phase.
018	P&ID Review	SSP to add stubouts on both sides of the Repressurization Tank so that a temporary tank can be installed.	1000-PID-0011	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
019	P&ID Review	SSP to add pumps to supernatant sump to allow for discharge to new clarifier.	1000-PID-0013	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	Pumps can be removed if plot is optimized in next phase such that supernatant sump can gravity drain.
020	P&ID Review	SSP to update abbreviation for sodium to Na rather than NA at all locations.	Global	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
021	P&ID Review	Motiva to consider DCS versus PLC control for demineralizers and polishers.	Global	Motiva	12/17/2025			Normal installation would be with a vendor provided PLC to control operation. DCS control allowing operator adjustments might void a performance guarantee.
022	P&ID Review	SSP to change depiction to show one mixed bed to avoid confusion on the number of mixed beds.	1000-PFD-0002	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
023	P&ID Review	Motiva to consider addition of a redundant caustic dilution exchanger.	1010-PID-0019	Motiva	12/17/2025			
024	P&ID Review	SSP to add double block and bleeds for all deaerator connections.	Global	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
025	P&ID Review	Motiva to consider addition of another BFW pump to provide N+2 redundancy.	1020-PID-0013	Motiva	12/17/2025			

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026	P&ID Review	Motiva to consider hard piping all calibration cylinders to totes.	Global	Motiva	12/17/2025			Top of calibration cylinder vent extends above top of tote. This may prevent a stacked tote configuration, which is the basis for FEL-2B.
027	P&ID Review	Motiva to consider changing control valves in fuel gas and boiler service to nitrogen rather than instrument air due to excess water in the instrument air system.	Global	Motiva	12/17/2025			
028	P&ID Review	Motiva to consider addition of sufficient instrumentation outside of boiler package to have triple redundancy on all instrumentation.	Global	Motiva	12/17/2025			
029	P&ID Review	Motiva to confirm where fuel gas and natural gas will be utilized in the boiler package.	1100-PID-0001	Motiva	12/17/2025			Motiva has stated potential concerns about total sulfur in flue gas as part of both Aromatics ISBL and the boiler complex.
030	P&ID Review	SSP to add knockout drum with bypass ahead of Fuel Gas Coalescer.	1100-PID-0001	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
031	P&ID Review	SSP to add redundant Fuel Gas Coalescer and remove bypass.	1100-PID-0001	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
032	P&ID Review	Motiva to consider installation of turbines on key pumps (e.g. demin and BFW) to get some benefit from HP steam produced.	Global	Motiva	12/17/2025			
033	P&ID Review	SSP to add double block and bleeds for all control valves in HP steam and BFW service.	Global	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
034	P&ID Review	SSP to add block and bleed outside of package limits between Boiler A and common HP steam header.	1100-PID-0013	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	Block valves are located near steam blow fitting on 1100-PID-0002. A bleed will be added between these blocks.
035	P&ID Review	Motiva to confirm environmental requirements for potential piping of fuel gas and natural gas bleeds to the flare.	1100-PID-0018	Motiva	12/17/2025			
036	P&ID Review	Motiva to consider adding requirement for backwash capability to lube oil exchanger in boiler bid package.	1100-PID-0020	Motiva	12/17/2025			
037	P&ID Review	Motiva to consider adding requirement for spare lube oil pump to boiler bid package	1100-PID-0020	Motiva	12/17/2025			
038	P&ID Review	SSP to add DCS indication to identify which boiler "D" phosphate pump is supplying.	1100-PID-0073	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
039	P&ID Review	SSP to add block valve and bleed to blowdown drum PSV.	1100-PID-0075	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
040	P&ID Review	SSP to install jumper on blowdown trim cooler to allow backwashing.	1100-PID-0079	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
041	P&ID Review	SSP to change back pressure regulator to full pressure control valve station in DCS with bypass.	1500-PID-0001	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
042	P&ID Review	Motiva to consider plate and frame exchanger for condensate application.	1500-PID-0007	Motiva	12/17/2025			
043	P&ID Review	SSP to check piping configuration on condensate trim cooler.	1500-PID-0011	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
044	P&ID Review	SSP to change cooling water exchanger backwash configuration to match Motiva's preferred method.	Global	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
045	P&ID Review	SSP to ensure that all exchangers have removable spools on channel side for maintenance.	Global	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
046	P&ID Review	SSP to ensure that cooling water supply enters on the bottom of exchangers.	Global	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
047	P&ID Review	Motiva to consider adding leak detection for ammonia tank unloading.	2000-PID-0001	Motiva	12/17/2025			
048	P&ID Review	SSP to add stubouts for temporary tanks to be installed to supply acid and caustic to demins during inspection.	2000-PID-0005 2000-PID-0007	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
049	P&ID Review	SSP to ensure gauge hatch is close to ladder/stairs on caustic tank so that tank can be gauged.	2000-PID-0005	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
050	P&ID Review	SSP to change sulfuric acid tank to drum.	2000-PID-0007	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
051	P&ID Review	Motiva to consider routing stormwater tank discharge (by others) to outfall, specifically 11 Reservoir.	General comment	Motiva	12/17/2025			
052	P&ID Review	Motiva to ensure a representative sample can be taken from the stormwater tank (by others).	General comment	Motiva	12/17/2025			

Client Name: Motiva
Project Title: Boiler Complex
Project Number: 25MA03

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SSP Engineering and Consulting
Date: 1/7/2026

Item	Requested By	Description	Associated P&ID	Action By	Requested Date	Required Date	Completion Date	Comments and Updates
053	P&ID Review	SSP to enclose oily water sump, add nitrogen blanket, and activated carbon on vent.	3000-PID-0005	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
054	P&ID Review	SSP to ensure that oily water hubs have water seals.	3000-PID-0005	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
055	P&ID Review	Motiva to determine if there is an environmental requirement to enclose oily water sump in an area that has no hydrocarbon.	3000-PID-0005	Motiva	12/17/2025			
056	P&ID Review	SSP to eliminate overflow on stormwater sump.	3000-PID-0001	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
057	P&ID Review	SSP to add diversion valves on high flow pumps to allow diversion of post first flush stormwater to 11 Reservoir.	3000-PID-0001	SSP	12/17/2025	1/16/2026	1/16/2026 (will be incorporated as part of FEL-2B report)	
058	P&ID Review	Motiva to consider addition of neutralization system on outlet of nonoily water sump.	3000-PID-0003	Motiva	12/17/2025			
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